Arul Rhik Mazumder

U.S. Citizen

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Research interests. I study what quantum algorithms really cost and when they are worth it: making algorithms such as Hamiltonian simulation cheaper on fault-tolerant hardware, proving how many T gates and ancilla qubits any circuit must use, and understanding when quantum approaches can genuinely outperform classical algorithms.

Education

Imperial College London **Sep. 2026 – Sep. 2027 (expected)**
MSc, Physics with Quantum Dynamics *London, UK*

Carnegie Mellon University **Aug. 2023 – May 2026**
B.S. in Computer Science; ML and Algorithms & Complexity concentrations *Pittsburgh, PA*

- Graduated with University Honors. Relevant coursework: Quantum Computing, Complexity Theory, Learning Theory, Randomized Algorithms, Convex Optimization, Parallel Computer Architecture.

Research Experience

Institute for Quantum Computing (IQC), University of Waterloo **May 2026 – Aug. 2026**
Undergraduate Researcher (Mitacs Globalink Scholar), hosted by Michele Mosca and Vlad Gheorghiu *Waterloo, ON, Canada*

- Developed tighter T-count bounds for structured unitaries: polynomial-time lower bounds for diagonal Clifford+T gates, computed from the gate's phase polynomial and exact for layers of disjoint Toffoli-type gates ($6m + 1$ T gates) in the $\{\text{CNOT}, T\}$ model with clean ancillas. Single-author preprint, arXiv:2610.01024; presented at QRE 2026 and UnitaryCon 2026.
- Paired the floor with an $O(n^3)$ upper bound that brackets every gate; the bounds carry over to Toffoli networks and finite-field multiplication oracles, recovering the known optimum of 17 T gates for the GF(4) multiplier.
- Completed USEQIP: quantum information coursework plus 30+ hours of hands-on experiments on IQC's quantum hardware.

Magic State Labs **Aug. 2026 – Oct. 2026**
Research Consultant, with Pradeep Niroula *Remote*

- Developed soundness rules for screening candidate problems for decoded quantum interferometry (DQI), and ran an AI-agent search under them that found two new candidate applications; written up as arXiv:2610.06753.

Caltech Institute for Quantum Information and Matter (IQIM) **May 2025 – Aug. 2025**
SURF Undergraduate Researcher, Preskill group, with Samson Wang and James Watson (Google Quantum AI) *Pasadena, CA*

- Worked on early fault-tolerant algorithms for matrix functions, using commutator scaling to lower the cost of product-formula routines; written up as arXiv:2608.13862 and presented as a poster at TQC 2026.
- Joined a biweekly reading group on quantum signal processing (QSP) and quantum singular value transformation (QSVT).

BlueQubit **Oct. 2025 – Present**
Part-time Researcher and Hackathon Lead, AI and Core teams *Los Angeles, CA (remote)*

- Benchmarked GPU-accelerated MPS and Pauli-path simulators against AWS Braket, Quantum Rings and open-source backends; the runtime scaling study was released as a preprint and informed product decisions and simulator cost estimators.
- Lead student and community engagement: turn BlueQubit research into accessible, technically accurate blog posts and tutorials, and mentor students and judge the BlueQubit track at student quantum hackathons.

Quantum Technologies Group, Carnegie Mellon University **May 2023 – Jun. 2025**
Researcher with Sridhar Tayur *Pittsburgh, PA*

- Wrote a tutorial paper with five accompanying notebooks on posing and solving QUBO problems with QAOA and quantum annealing (*Tutorials in Operations Research*, INFORMS 2025, pp. 145–183).
- Implemented and tested semidefinite-programming formulations for entanglement detection using CVXPY, PICOS and Gurobi.

Algorithms Group, George Mason University **May 2022 – Jun. 2025**
Researcher with Fei Li *Remote*

- Implemented quantum string matching in Qiskit, benchmarked against Rabin–Karp and Knuth–Morris–Pratt (ACM AISS 2022).
- Built a hybrid quantum-classical algorithm for the traveling salesman problem on D-Wave QPUs that outperformed the pure classical and pure quantum approaches it was benchmarked against (IEEE IPDPS 2023).
- Designed three classical heuristics and a D-Wave hybrid algorithm for shipment rerouting, tested with up to 1,000 shipments on real transport networks (Chicago, Barcelona, Winnipeg) against classical solvers (JSSR 2024).

Publications

Manuscripts and preprints

1. **A. R. Mazumder**. "Exact T-counts of CCZ Layers from an Isotropy Bound." arXiv:2610.01024, 2026. Proves that Hadamard-free circuits for a layer of m disjoint CCZ gates need $6m + 1$ T gates, matching Campbell and Howard's construction.
2. **A. R. Mazumder**, P. Niroula. "Discovering New Problems for Decoded Quantum Interferometry." arXiv:2610.06753, 2026.
3. **A. R. Mazumder**, J. Watson, S. Wang. "Resource-Efficient Quantum Eigenvalue Transform with Commutator Scaling." arXiv:2608.13862; *Theory of Quantum Computation, Communication and Cryptography (TQC)*, 2026, poster.
4. **A. R. Mazumder**. "Ancilla-free Randomized Toffoli Gates and the T-count of Boolean Oracles." arXiv preprint (submitted), 2026. Randomized Toffoli gates with no extra qubits, within a factor of eight of optimal for measurement-free circuits.
5. **A. R. Mazumder**, H. Karapeytan, M. Z. Mullath, R. Tanin, H. Gharibyan, H. Tepanyan. "Benchmarking Zero-Setup Quantum Circuit Simulators." arXiv:2607.09882, 2026. Runtime scaling study of GPU-accelerated circuit simulators.
6. F. Li, **A. R. Mazumder**, M. Zhao. "Quantum Annealing Approaches to Shipment Rerouting Problems." arXiv:2501.05624, 2025.

Peer-reviewed

1. **A. R. Mazumder**, E. Towe. "An End-to-End Hybrid Quantum–Classical Sampling Workflow for Discrete Markov Random Fields." *IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2026; arXiv:2607.09893.
2. **A. R. Mazumder**, S. R. Mazumder. "Depth-Efficient Quantum Topological Data Analysis for Regime-Specific Detection of Financial Stress." *IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2026; arXiv:2607.09906.
3. **A. R. Mazumder**, S. Tayur. "Five Starter Problems: Solving Quadratic Unconstrained Binary Optimization Models on Quantum Computers." *Tutorials in Operations Research*, INFORMS, 2025, pp. 145–183.
4. M. Zhao, **A. R. Mazumder**, F. Li. "Improved Classical and Quantum Algorithms for the Shipment Rerouting Problems." *Journal of Student-Scientists' Research*, 2024, doi:10.13021/jssr2024.4263.
5. **A. R. Mazumder**, A. Sen, U. Sen. "Benchmarking Metaheuristic-Integrated QAOA Against Quantum Annealing." *LNNS*, 2024.
6. F. Li, **A. R. Mazumder**. "An Adaptive Hybrid Quantum Algorithm for the Metric Traveling Salesman Problem." *IPDPS* 2023.
7. M. Gao, R. Huang, **A. R. Mazumder**, F. Li. "Comparisons of Classic and Quantum String Matching Algorithms." *AISS* 2022.

Talks and Posters

- **UnitaryCon 2026**, Toronto — poster, "Certified T-count floors for Toffoli networks and reversible circuits."
- **Quantum Resource Estimation (QRE) 2026**, Toronto — poster, same title.
- **IEEE QCE 2026**, Toronto — two first-author papers.
- **TQC 2026** — poster, "Resource-Efficient Quantum Eigenvalue Transform with Commutator Scaling" (PDF).
- **Quantum Leap Career Nexus**, University of Maryland, Nov. 2025, and **Caltech IQIM SURF Seminar**, Aug. 2025 — talks, "Early Fault-Tolerant Quantum Algorithms for Matrix Functions via Trotter Extrapolation" (slides).
- **INFORMS Annual Meeting 2025** — presentation on solving QUBO models on quantum computers.

Awards and Honors

- **IEEE QCE Student Travel Grant** (2026), IEEE Quantum Technical Community.
- **Mitacs Globalink Research Award and IQC Undergraduate Research Award** (2026), funding the IQC research term.
- **Duke STAQ NSF-funded Invited Scholar** (2025): week-long PhD-level seminar on quantum simulation, error correction and architecture with Peter Love, Kenneth Brown and Yu Tong, including a Duke Quantum Center lab tour.
- **MIT iQuHACK** (2026): 2nd, NVIDIA track, of 90+ teams; variational LABS with Pauli correlation encodings on GPUs.
- **YQuantum** (2025, Yale): Yale Quantum Institute Grand Prize; 1st place, Alice & Bob track; 2nd overall; Wigner-function quantum state reconstruction via convex optimization and convolutional-network denoising.
- **Citadel/Correlation One Datathon** (2024): 2nd of 150 invited from 3,000+ applicants; pairs trading, Sharpe 1.58.
- **HackCMU** (2023): Graduate Student Association Infrastructure Prize for Eco-Bin, an AI waste sorter piloted with CMU Dining.
- **Regeneron Science Talent Search Scholar** (2023), top 300 of 2,000+ applicants nationally.
- **Olympiads:** USA(J)MO qualifier, USAPhO Bronze, USACO Gold, USNCO National Finalist.

Leadership

CMU Quantum — **Sep. 2024 – May 2026**
Founder and President — *Pittsburgh, PA*

- Founded CMU's first student quantum organization and grew it to 200+ members and 10+ student leaders; secured sponsorship from General Motors and Boeing; ran workshops with IBM, IonQ and Quantinuum.
- Developed teaching materials (slides and handouts), led hackathon tracks, and organized sponsored research trips.

Technical Skills

Languages and proof tools: Python, Julia, C++, Lean 4
Quantum and optimization: Qiskit, PennyLane, CUDA-Q, D-Wave Ocean; CVXPY, PICOS, Gurobi; CUDA, MPI